

$$= ABC'.1 + AB'C'.1 + CD'B.1 + C'D'B'.1 + A'CD'.1 + DC'AB + DC'AB' + DBA'C + DBAC' \text{ (Postulate 2(b))}$$

$$= ABC'.(D+D') + AB'C'.(D+D') + CD'B.(A+A') + C'D'B'.(A+A') + A'CD'.(B+B') + DC'AB + DC'A'B' + DBA'C + DBAC' \text{ (Postulate 5(a))}$$

$$= \overline{ABC'D} + \overline{AB'C'D} + \overline{AB'C'D} + \overline{AB'C'D} + \overline{C'D'BA} + \overline{C'D'BA} + \overline{C'D'BA} + \overline{C'D'BA} + \overline{A'CD'B} + \overline{A'CD'B} + \overline{DC'AB} + \overline{DC'A'B'} + \overline{DBA'C} + \overline{DBAC'} \text{ (Postulate 4(a))}$$

Rearranging and removing repeated terms:

$$\overline{A'B'C'D} + \overline{A'B'C'D} + \overline{A'B'C'D} + \overline{A'B'C'D} + \overline{A'BCD} + \overline{A'BCD} + \overline{A'BCD} + \overline{A'BCD} + \overline{AB'C'D} + \overline{AB'C'D} + \overline{AB'C'D} + \overline{AB'C'D}$$

$$\overline{AB'CD} + \overline{AB'CD} + \overline{AB'CD} + \overline{AB'CD}$$

$$\Sigma(0,1,2,4,6,7,8,9,10,11,12,13) = F$$

$$F' = \Sigma(3,5,14,15)$$

$$= ABC + AB'CD + A'BCD$$

$$F^0 = (F')' = (ABC + AB'CD + A'BCD)'$$

$$= (ABC)' \cdot (AB'CD)' \cdot (A'BCD)' \text{ (Extended Thm 5(a))}$$

$$= (A+B+C)' \cdot (A''+B''+C''+D')' \cdot (A''+B'+C''+D')' \text{ (Extended Thm 5(b))}$$

$$= (A+B+C)' \cdot (A+B+C+D)' \cdot (A+B+C+D)' \text{ (Thm 3)}$$

	CD 00	01	11	10
AB 00			1	
01		1		
11			1	1
10				

